MUKESH PATEL SCHOOL OF TECHNOLOGY MANAGEMENT & ENGINEERING

Shashwat Jain

BTech Artificial Intelligence & Machine Learning

2027 Batch | Date of Birth: 5th October 2005

Portfolio: maishashwathu.github.io/Portfolio/

OBJECTIVE

B.Tech AI & ML student seeking an AI/ML engineering role, with hands-on experience in document-grounded RAG, deep learning and LLM quantization. Focused on building practical AI systems and evaluating model quality, memory use and inference efficiency.

EDUCATION

SVKM'S NMIMS MPSTME, Shirpur Jul 2023 - Present

B.Tech, Artificial Intelligence and Machine Learning | **CGPA: 3.41/4.0** | Shirpur

Mercy Memorial School Apr 2022 - Apr 2023

Class XII - ICSE | **Percentage: 80%**

Mercy Memorial School Apr 2020 - Apr 2021

Class X - ICSE | **Percentage: 88.5%**

MAJOR PROJECTS

LLM Post-Training Quantization | Capstone

Python, PyTorch, CUDA, Triton, NVML

- Benchmarked **FP4, NF4 and NVFP4** against 16-bit baselines across **five LLM families**: Mistral, Llama, Qwen, Phi and Gemma.
- Evaluated quality, GPU memory, latency, throughput and GPU energy; profiled quantized execution paths and repeated probes to assess workload-dependent trade-offs.

Multilingual RAG Platform

Sep 2025 - Nov 2025

LangChain, FAISS, RAG, Prompt Engineering

- Built a multilingual chatbot that retrieves document context with **FAISS** and generates document-grounded responses through LangChain.
- Implemented document chunking, embeddings, query classification and prompt engineering to support relevant retrieval and grounded answer generation.

VoxCeleb Speaker Identification

Aug 2025 - Oct 2025

PyTorch, CNN, Spectrogram Augmentation

- Built an end-to-end speaker identification system on VoxCeleb, achieving **86.6% validation accuracy**.
- Designed a VGG16-Lite-inspired 2D CNN with batch normalization, dropout and spectrogram augmentation for speaker classification.

ESG Risk Analysis Dashboard

Power BI, Data Modeling, Interactive Visualization

- Built **English and Marathi Power BI dashboards** to analyze village-level ESG survey results, risk indices and environmental, social and governance scores.
- Enabled filtering by village, focus group and ESG dimension, with question-level analysis and response counts to investigate sources of risk.

SKILLS

Generative AI: RAG, LangChain, FAISS, Prompt Engineering, Post-Training Quantization

Machine Learning & DL: PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy

Programming & Databases: Python, C++, MySQL, MongoDB

Tools & Visualization: Git, GitHub, Streamlit, FastAPI, Power BI, Tableau

Operating Systems: Windows, Linux | **Other:** AutoCAD, TinkerCAD, Cisco Packet Tracer, Blender

CERTIFICATION

- Deep Learning - IIT Ropar (NPTEL, Elite Category)
- Machine Learning Using Python - NPTEL (SWAYAM)

ACHIEVEMENTS & RESPONSIBILITIES

- **Innovent 2025:** Participant, District-Level Idea Competition, NMIMS
- **HackRx 6.0 Hackathon (2025):** Reached Round 4 (Pre-Finals)
- **ANRF AISEHack 2.0 (2026):** Attended the final round and stood 4th in the Kaggle leaderboard.
- **Vice Chair, ACM Student Chapter:** Contributed to technical events, knowledge-sharing sessions and student-led initiatives.